Тетрахлорид титана

Тетрахлорид титана — это неорганическое соединение с формулой $4Cl_{Ti}$. Это важное промежуточное соединение при производстве металлического титана и пигмента диоксида титана. $TiCl_4$ — это летучая жидкость. При контакте с влажным воздухом она образует густые облака диоксида титана (TiO_2) и соляной кислотой — реакция, которая раньше использовалась в дымовых машинах. Иногда ее называют «тикл» или «тикл 4» — по звучанию символов ее молекулярной формулы ($TiCl_4$).[7][8]



Crystals of frozen titanium tetrachloride melting into the liquid

Properties and structure

$TiCl_4$ is a dense, colourless liquid, although crude samples may be yellow or even red-brown. It is one of the rare transition metal halides that is a liquid at room temperature, VCl_4 being another example. This property reflects the fact that molecules of $TiCl_4$ weakly self-associate. Most metal chlorides are polymers, wherein the chloride atoms bridge between the metals. Its melting point is similar to that of CCl_4.[9][10]

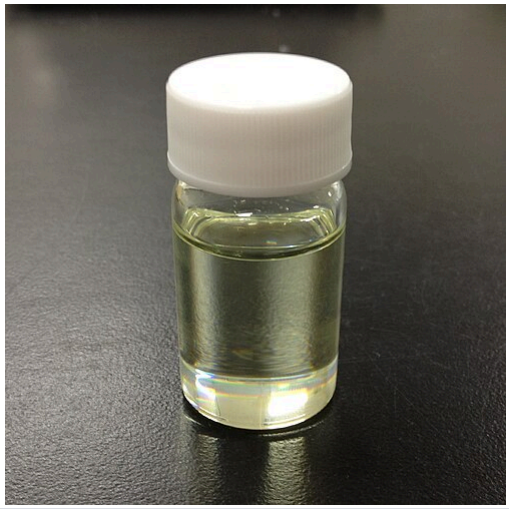
Ti^{4+} has a "closed" electronic shell, with the same number of electrons as the noble gas argon. The tetrahedral structure for $TiCl_4$ is consistent with its description as a d^0 metal center (Ti^{4+}) surrounded by four identical ligands. This configuration leads to highly symmetrical structures, hence the tetrahedral shape of the molecule. $TiCl_4$ adopts similar structures to $TiBr_4$ and TiI_4; the three compounds share many similarities. $TiCl_4$ and $TiBr_4$ react to give mixed halides $TiCl_{4-x}Br_x$, where x = 0, 1, 2, 3, 4. Magnetic resonance measurements also indicate that halide exchange is also rapid between $TiCl_4$ and VCl_4.[11]

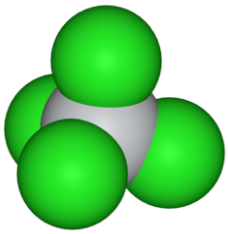
$TiCl_4$ is soluble in toluene and chlorocarbons. Certain arenes form complexes of the type $[(C_6R_6)TiCl_3]^+$.[12] $TiCl_4$ reacts exothermically with donor solvents such as THF to give hexacoordinated adducts.[13] Bulkier ligands (L) give pentacoordinated adducts $TiCl_4L$.

Production

$TiCl_4$ is produced by the chloride process, which involves the reduction of titanium oxide ores, typically ilmenite ($FeTiO_3$), with carbon under flowing chlorine at 900 °C. Impurities are removed by distillation.[10]

Тетрахлорид титана





Имена

Название IUPAC
Хлорид титана (IV)

Другие названия
Тетрахлорид титана
Тетрахлоротитан

Идентификаторы

Номер CAS	7550-45-0 (https://commonchemistry.cas.org/detail?cas_rn=7550-45-0) ✓
3D-модель (JSmol)	Интерактивное изображение (https://chemapps.stolaf.edu/jmol/jmol.php?model=Cl%5BTi%5D%28Cl%29%28Cl%29Cl)
Чеби	ЧЕБИ:231499 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=231499)

Химспидер	22615 (https://www.chemspider.com/Chemical-Structure.22615.html) ✓
Информационная карта ECHA	100.028.584 (https://echa.europa.eu/substance-information/-/substanceinfo/100.028.584)
Номер EC	231-441-9
Сетка	Титан + тетрахлорид (https://www.nlm.nih.gov/cgi/mesh/2014/MB_cgi?mode=&term=Titanium+tetrachloride)
PubChem CID	24193 (https://pubchem.ncbi.nlm.nih.gov/compound/24193)
Номер RTECS	XR1925000
UNII	8O3PJE5T7Q (https://precision.fda.gov/uniisearch/srs/unii/8O3PJE5T7Q) ✓
Номер ООН	1838
Панель управления CompTox (EPA)	DTXSID8042476 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID8042476)
Инчи	
УЛЫБАЕТСЯ	
Свойства	
Химическая формула	$TiCl_4$
Молярная масса	189,67 г·моль$^{-1}$

Внешний вид	Бесцветная жидкость
Запах	проникающий кислотный запах
Плотность	1,726 г/см^3
Температура плавления	−24,1 °C (−11,4 °F; 249,1 K)
Температура кипения	136,4 °C (277,5 °F; 409,5 K)
Растворимость в воде	вступает в реакцию (экзотермический гидролиз)[1]
Растворимость	растворим в дихлорметане,[2] толуоле,[3] пентане[4]
Давление пара	1,3 кПа (20 °C)
Магнитная восприимчивость (χ)	−54,0·10^{-6} см^3/моль
Показатель преломления (n_D)	1,61 (10,5 °C)
Вязкость	827 мкПа·с
Структура	
Координационная геометрия	Тетрагональный
Молекулярная форма	Четырехгранный
Дипольный момент	0 D
Термохимия	
Стандартная молярная энтропия ($S^\ominus_{298}$)	355 Дж·моль$^{-1}$·К$^{-1}$[5]
Стандартная энтальпия образования ($\Delta_f H^\ominus_{298}$)	−763 кДж·моль$^{-1}$[5]

Опасности[6]	
Охрана труда (ОТ):	
Основные опасности	Токсичен, вызывает коррозию, вступает в реакцию с водой с выделением HCl
Маркировка GHS:	
Пиктограммы	
Сигнальное слово	**Опасность**
Указания об опасности	H314, H317, H330, H335, H370, H372
Меры предосторожности	P280, P301+P330+P331, P304+P340, P305+P351+P338, P308+P310
NFPA 704 (пожарный бриллиант)	
Паспорт безопасности (SDS)	MSDS (http://www.sciencelab.com/msds.php?msdsId=9925270)
Родственные соединения	
Другие анионы	Бромид титана(IV) Фторид титана(IV) Иодид титана(IV)
Другие катионы	Хлорид гафния(IV) Хлорид циркония(IV)
Родственные соединения	Хлорид титана(II) Хлорид титана(III)

Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).

✓ verify (what is ✓✗ ?)

Infobox references

$$2\ FeTiO_3 + 7\ Cl_2 + 6\ C \rightarrow 2\ TiCl_4 + 2\ FeCl_3 + 6\ CO$$

The coproduction of $FeCl_3$ is undesirable, which has motivated the development of alternative technologies. Instead of directly using ilmenite, "rutile slag" is used. This material, an impure form of TiO_2, is derived from ilmenite by removal of iron, either using carbon reduction or extraction with sulfuric acid. Crude $TiCl_4$ contains a variety of other volatile halides, including vanadyl chloride ($VOCl_3$), silicon tetrachloride ($SiCl_4$), and tin tetrachloride ($SnCl_4$), which must be separated.[10]

Applications

Production of titanium metal

The world's supply of titanium metal, about 250,000 tons per year, is made from $TiCl_4$. The conversion involves the reduction of the tetrachloride with magnesium metal. This procedure is known as the Kroll process:[14]

$$2\ Mg + TiCl_4 \rightarrow 2\ MgCl_2 + Ti$$

In the Hunter process, liquid sodium is the reducing agent instead of magnesium.[15]

Production of titanium dioxide

Around 90% of the $TiCl_4$ production is used to make the pigment titanium dioxide (TiO_2). The conversion involves hydrolysis of $TiCl_4$, a process that forms hydrogen chloride:[14]

$$TiCl_4 + 2\ H_2O \rightarrow TiO_2 + 4\ HCl$$

In some cases, $TiCl_4$ is oxidised directly with oxygen:

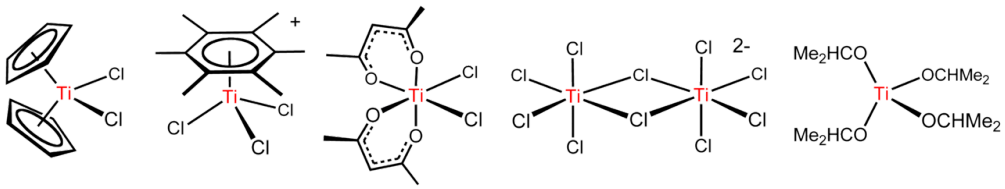
$$TiCl_4 + O_2 \rightarrow TiO_2 + 2\ Cl_2$$

Smoke screens

It has been used to produce smoke screens since it produces a heavy, white smoke that has little tendency to rise. "Tickle" was the standard means of producing on-set smoke effects for motion pictures, before being phased out in the 1980s due to concerns about hydrated HCl's effects on the respiratory system.

Chemical reactions

Titanium tetrachloride is a versatile reagent that forms diverse derivatives including those illustrated below.[16]



Alcoholysis and related reactions

A characteristic reaction of $TiCl_4$ is its easy hydrolysis, signaled by the release of HCl vapors and titanium oxides and oxychlorides. Titanium tetrachloride has been used to create naval smokescreens, as the hydrochloric acid aerosol and titanium dioxide that is formed scatter light very efficiently. This smoke is corrosive, however.[10]

Alcohols react with $TiCl_4$ to give alkoxides with the formula $[Ti(OR)_4]_n$ (R = alkyl, n = 1, 2, 4). As indicated by their formula, these alkoxides can adopt complex structures ranging from monomers to tetramers. Such compounds are useful in materials science as well as organic synthesis. A well known derivative is titanium isopropoxide, which is a monomer. Titanium bis(acetylacetonate)dichloride results from treatment of titanium tetrachloride with excess acetylacetone:[17]

$$TiCl_4 + 2\ Hacac \rightarrow Ti(acac)_2Cl_2 + 2\ HCl$$

Organic amines react with $TiCl_4$ to give complexes containing amido (R_2N^--containing) and imido (RN^{2-}-containing) complexes. With ammonia, titanium nitride is formed. An illustrative reaction is the synthesis of tetrakis(dimethylamido)titanium $Ti(N(CH_3)_2)_4$, a yellow, benzene-soluble liquid:[18] This molecule is tetrahedral, with planar nitrogen centers.[19]

$$4\ LiN(CH_3)_2 + TiCl_4 \rightarrow 4\ LiCl + Ti(N(CH_3)_2)_4$$

Complexes with simple ligands

$TiCl_4$ is a Lewis acid as implicated by its tendency to hydrolyze. With the ether THF, $TiCl_4$ reacts to give yellow crystals of $TiCl_4(THF)_2$. With chloride salts, $TiCl_4$ reacts to form sequentially $[Ti_2Cl_9]^-$, $[Ti_2Cl_{10}]^{2-}$ (see figure above), and $[TiCl_6]^{2-}$.[20] The reaction of chloride ions with $TiCl_4$ depends on the counterion. $[N(CH_2CH_2CH_2CH_3)_4]Cl$ and $TiCl_4$ gives the pentacoordinate complex $[N(CH_2CH_2CH_2CH_3)_4][TiCl_5]$, whereas smaller $[N(CH_2CH_3)_4]^+$ gives $[N(CH_2CH_3)_4]_2[Ti_2Cl_{10}]$. These reactions highlight the influence of electrostatics on the structures of compounds with highly ionic bonding.

Redox

Reduction of $TiCl_4$ with aluminium results in one-electron reduction. The trichloride ($TiCl_3$) and tetrachloride have contrasting properties: the trichloride is a colored solid, being a coordination polymer, and is paramagnetic. When the reduction is conducted in THF solution, the Ti(III) product converts to the light-blue adduct $TiCl_3(THF)_3$.

Organometallic chemistry

The organometallic chemistry of titanium typically starts from $TiCl_4$. An important reaction involves sodium cyclopentadienyl to give titanocene dichloride, $TiCl_2(C_5H_5)_2$. This compound and many of its derivatives are precursors to Ziegler–Natta catalysts. Tebbe's reagent, useful in organic chemistry, is an aluminium-containing derivative of titanocene that arises from the reaction of titanocene dichloride with trimethylaluminium. It is used for the "olefination" reactions.[16]

Arenes, such as $C_6(CH_3)_6$ react to give the piano-stool complexes $[Ti(C_6R_6)Cl_3]^+$ (R = H, CH_3; see figure above). This reaction illustrates the high Lewis acidity of the $TiCl_3^+$ entity, which is generated by abstraction of chloride from $TiCl_4$ by $AlCl_3$.[12]

Reagent in organic synthesis

$TiCl_4$ finds occasional use in organic synthesis, capitalizing on its Lewis acidity, its oxophilicity, and the electron-transfer properties of its reduced titanium halides. It is used in the Lewis acid catalysed aldol addition[21] Key to this application is the tendency of $TiCl_4$ to activate aldehydes (RCHO) by formation of adducts such as $(RCHO)TiCl_4OC(H)R$.[22]

Toxicity and safety considerations

Hazards posed by titanium tetrachloride generally arise from its reaction with water that releases hydrochloric acid, which is severely corrosive itself and whose vapors are also extremely irritating. $TiCl_4$ is a strong Lewis acid, which exothermically forms adducts with even weak bases such as THF and water.

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General reading

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External links

- Titanium tetrachloride: Health Hazard Information (https://web.archive.org/web/20011202123121/http://www.epa.gov/ttn/atw/hlthef/titanium.html)
- NIST Standard Reference Database (http://webbook.nist.gov/chemistry/)
- ChemSub Online: Titanium tetrachloride (http://chemsub.online.fr/name/titanium_tetrachloride)